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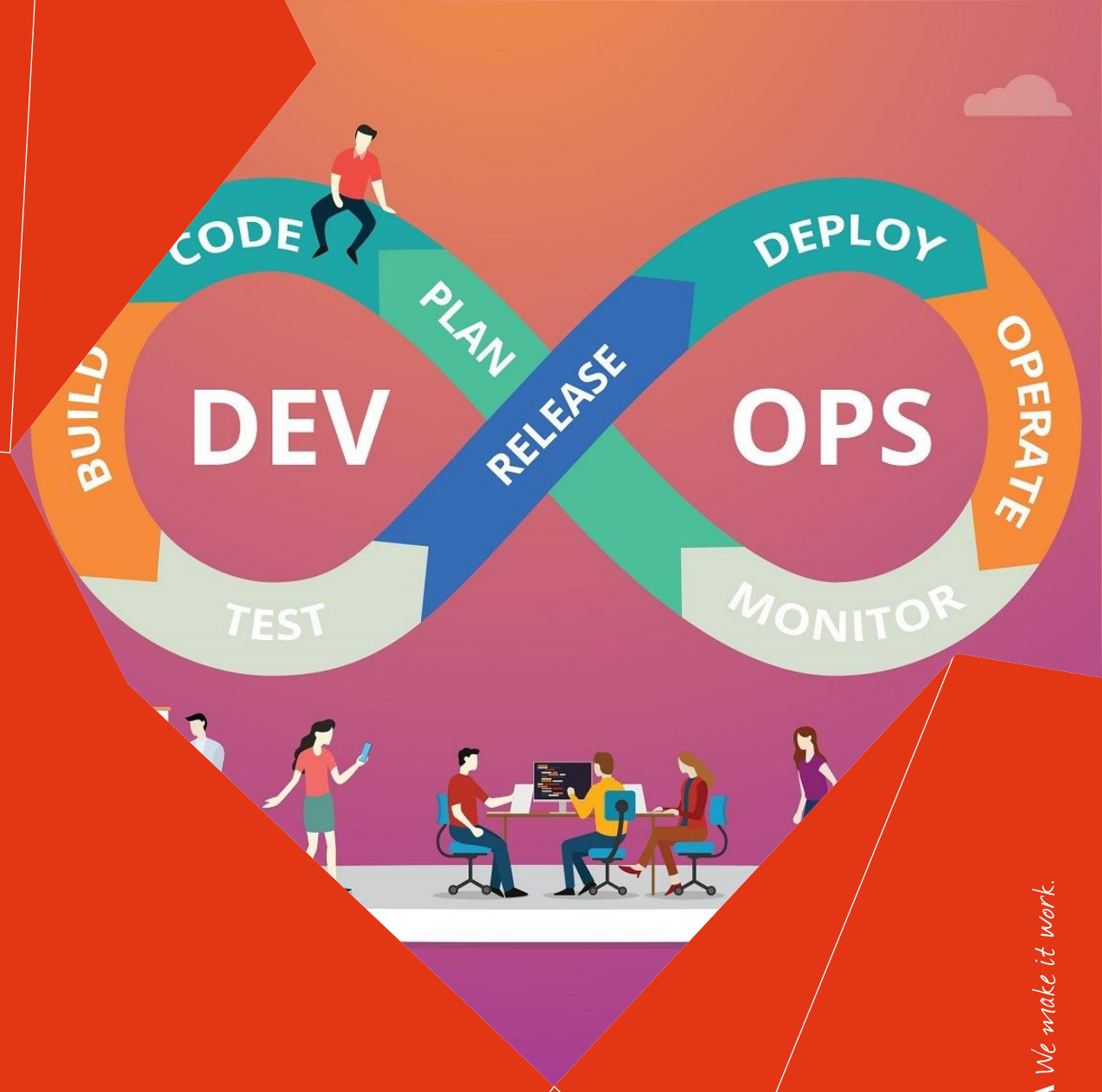
08/11/2025

Platform Engineering

BATbern54

Michele Monforte / Fabian Nicollier



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Introduction	Michele	10’
Self-service	Michele	5’
Use case (Support process)	Michele	5’
Landing Zones / Modules	Fabian	5’
Accelerators	Fabian	5’
Q/A		

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Purpose of Platform engineering

Systems are becoming rapidly more complex

They can only really be operated by the people who write them

And you can't do a good job of writing them unless you are regularly exposed to the feedback loops of operating them

Is the Platform Engineering team responsible for SLOs, service uptime, and a reliable customer experience ? **NO**

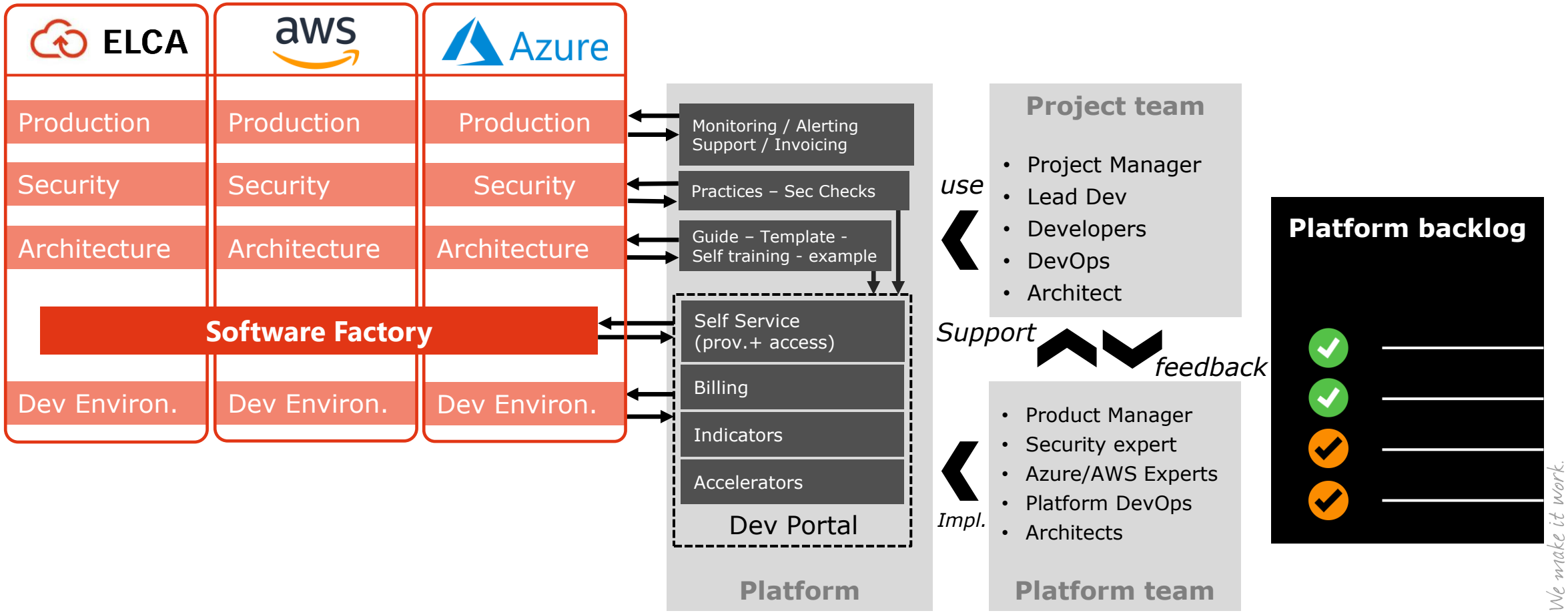
Is Time to spin up new service or db or other developer workflow 1st priority ? **YES**

Improve **Dev**eloper **Ex**perience by providing :

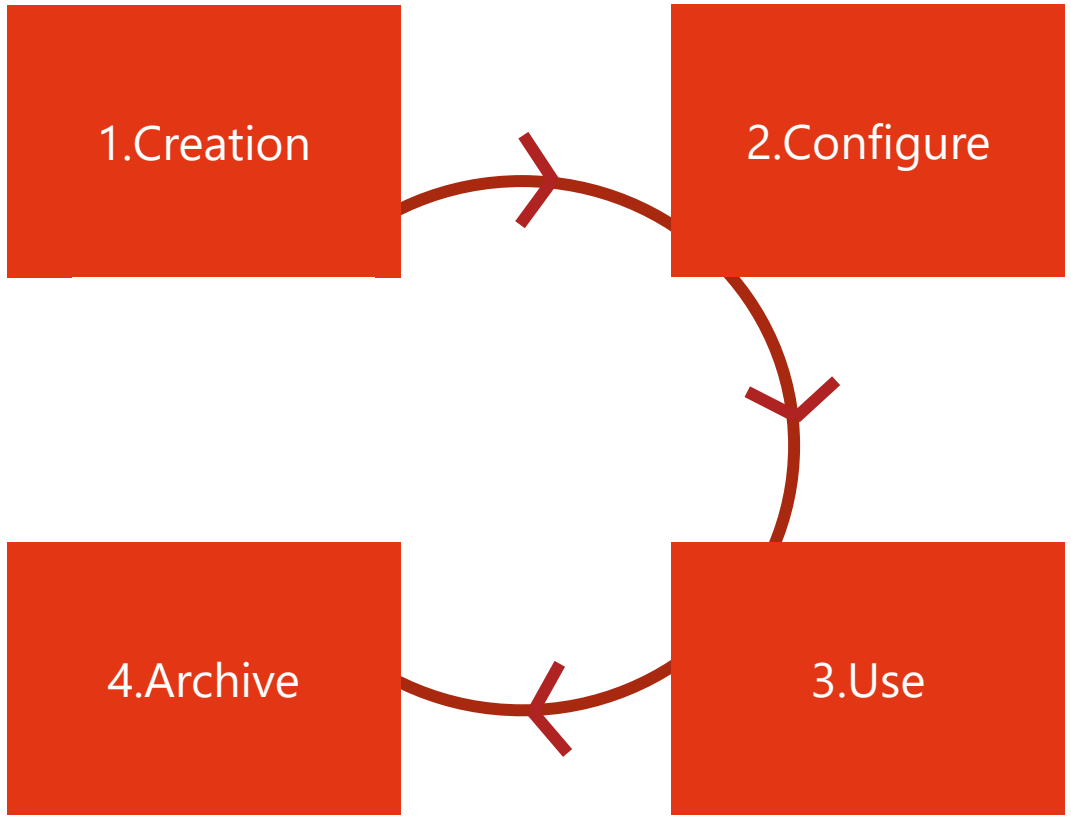
- All information possible to improve code.
- All information to make better technical choices.
- Reduce time to bootstrap technologies.
- Implement DevOps – Developer ownership

The future of Ops is platform engineering | PlatformCon 2023

Platform as a product



Asset Lifecycle



Example

1. Order an Azure Subscription
2. Create/Renew a Service Principal
3. Check your Azure consumption
4. Archive your Account

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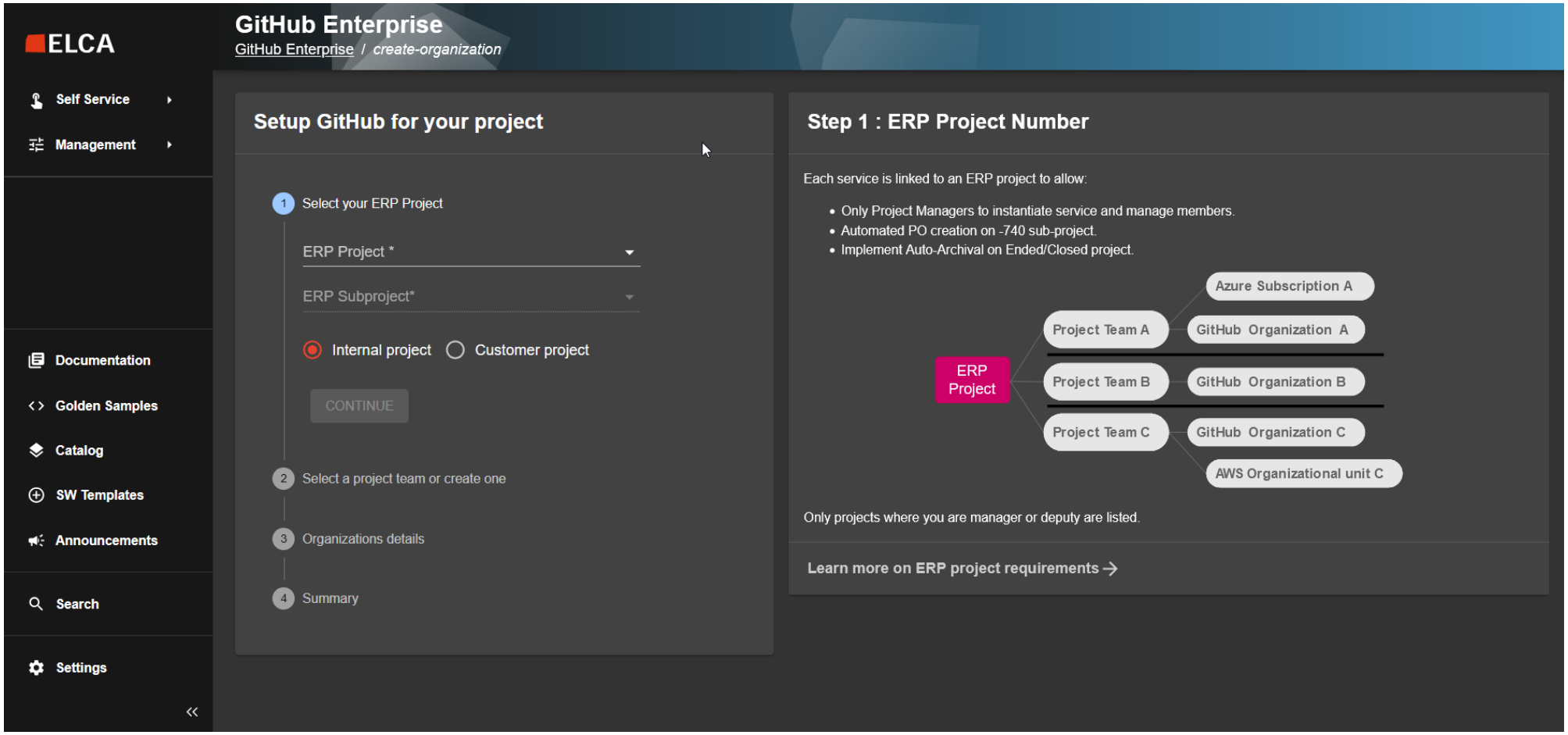
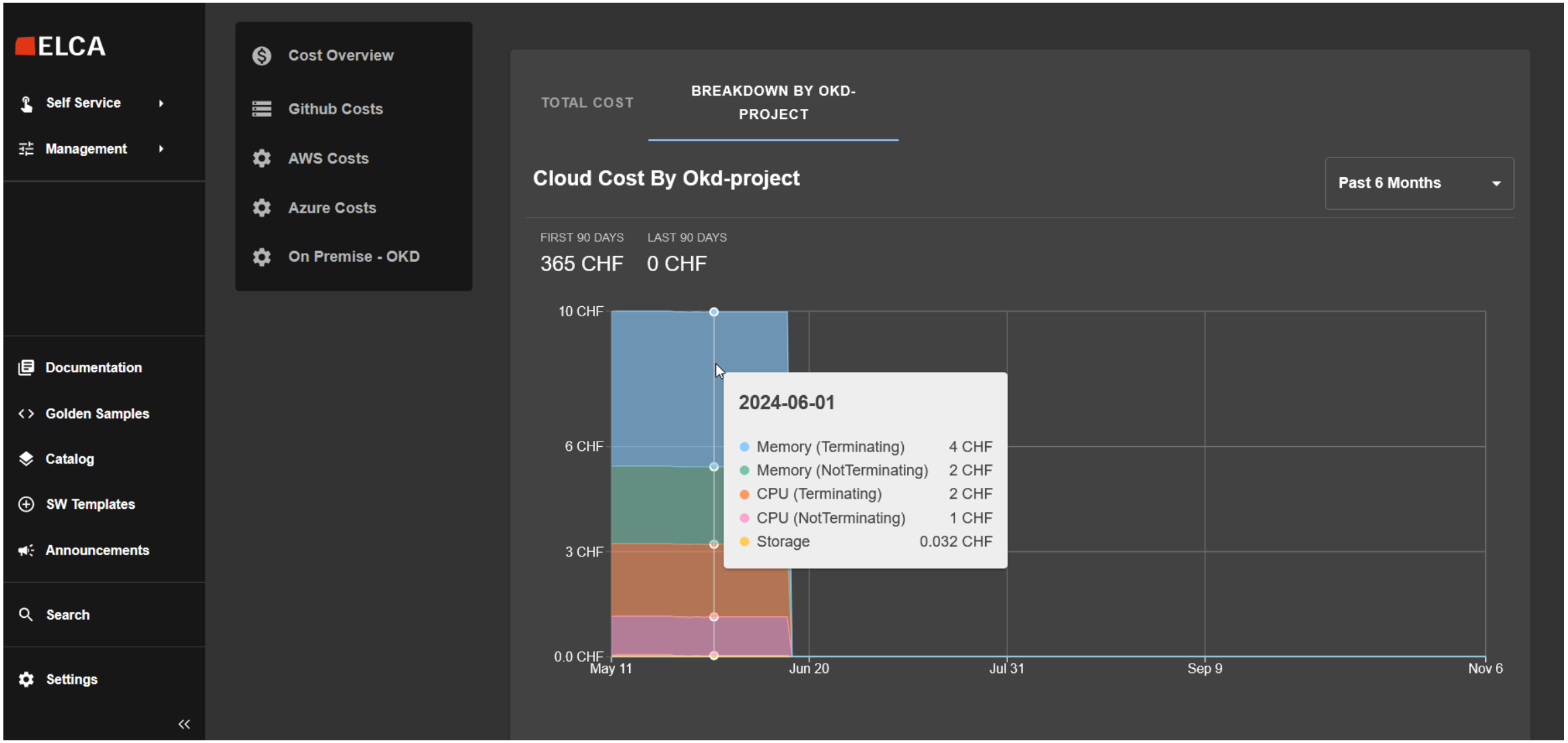
Self-Service

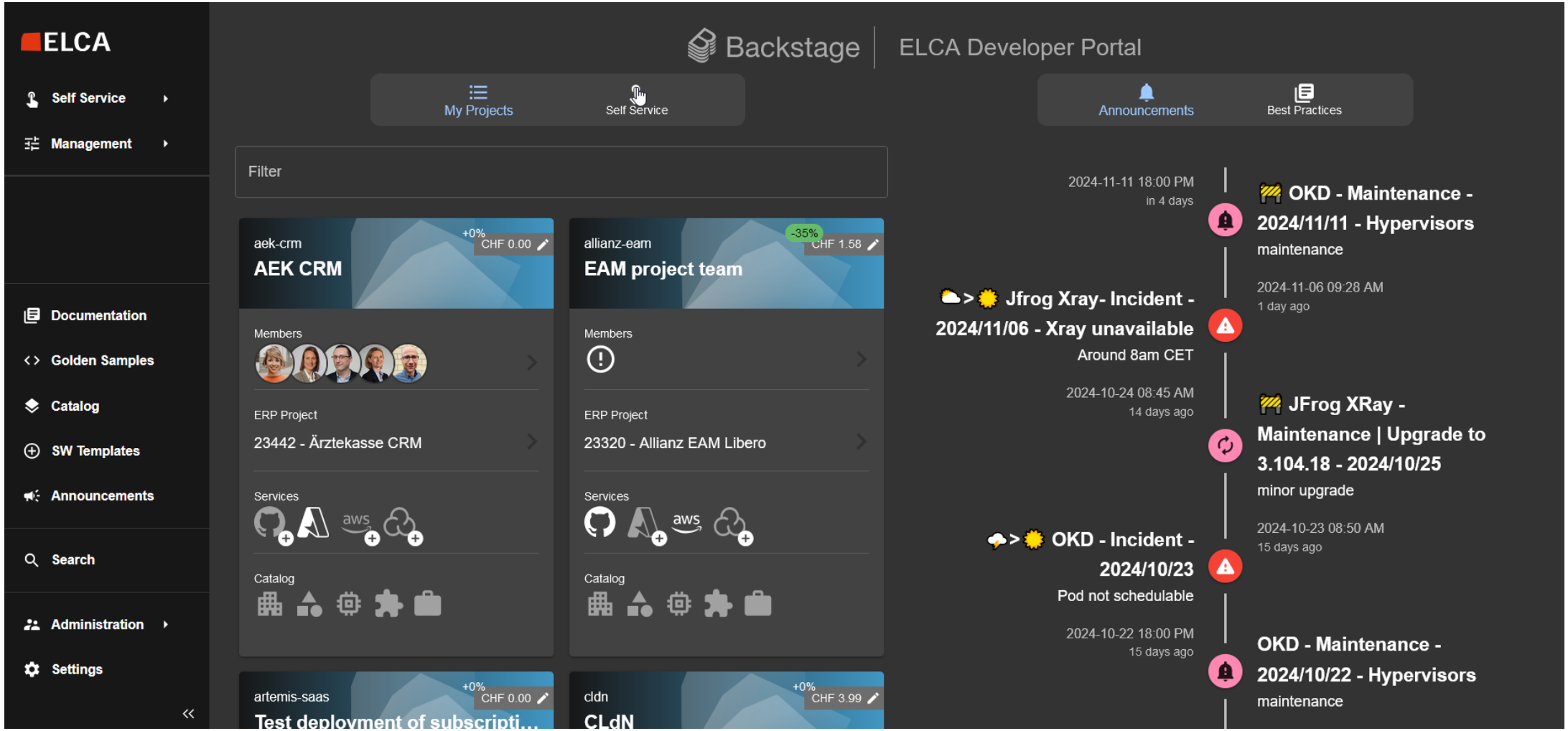


Self-Service : Integration with ERP, Entity location, Project Organisation mapping

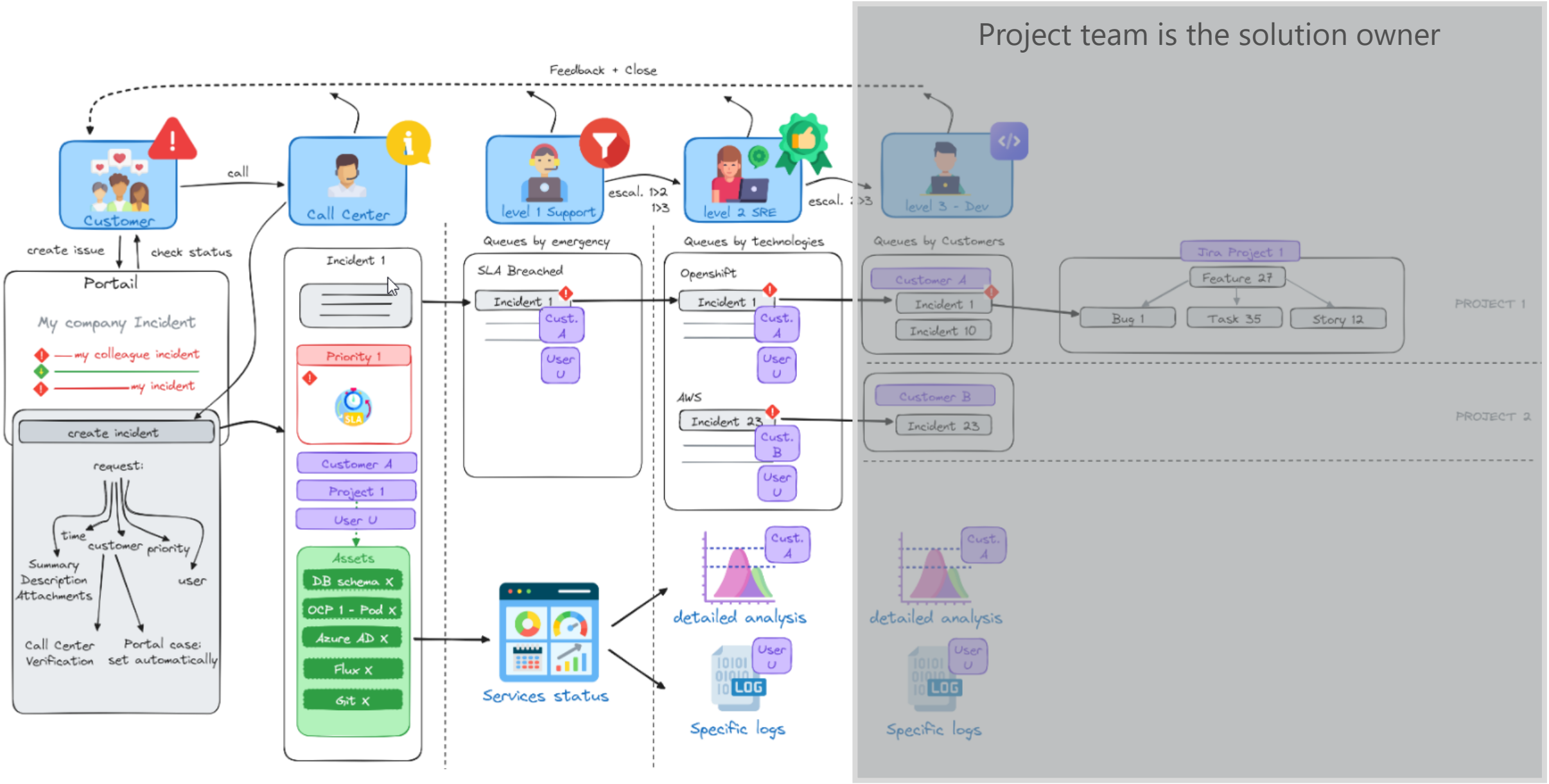
Cost Allocation and Consumption



Communication – Dashboard



Use Case – Support Process



Landing Zone Building Blocks

Landing Zone

- Highly Modular
- Establish Governance
- Non-Productive or Productive

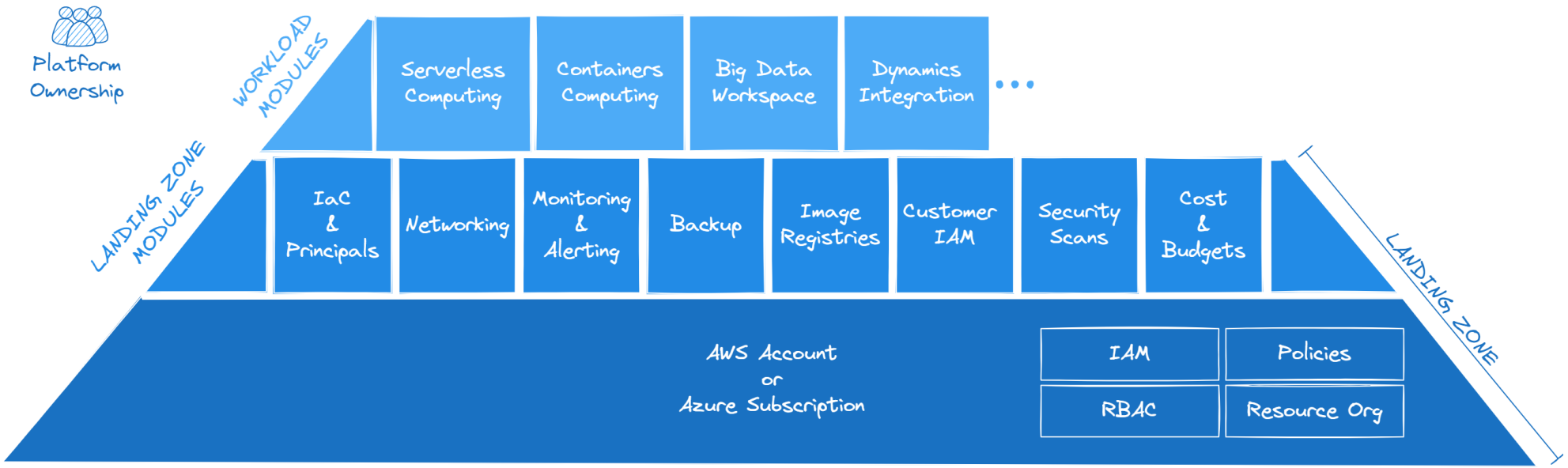
Module

e.g. Networking

- Modules are interdependent
- Parameters provide a "Generic" configuration
- Delivered in different ways

Procedure / Guideline
Infra Code Snippet
Versioned & Shared Module
Provisioned via Backstage

+MATURE
+COSTLY $$



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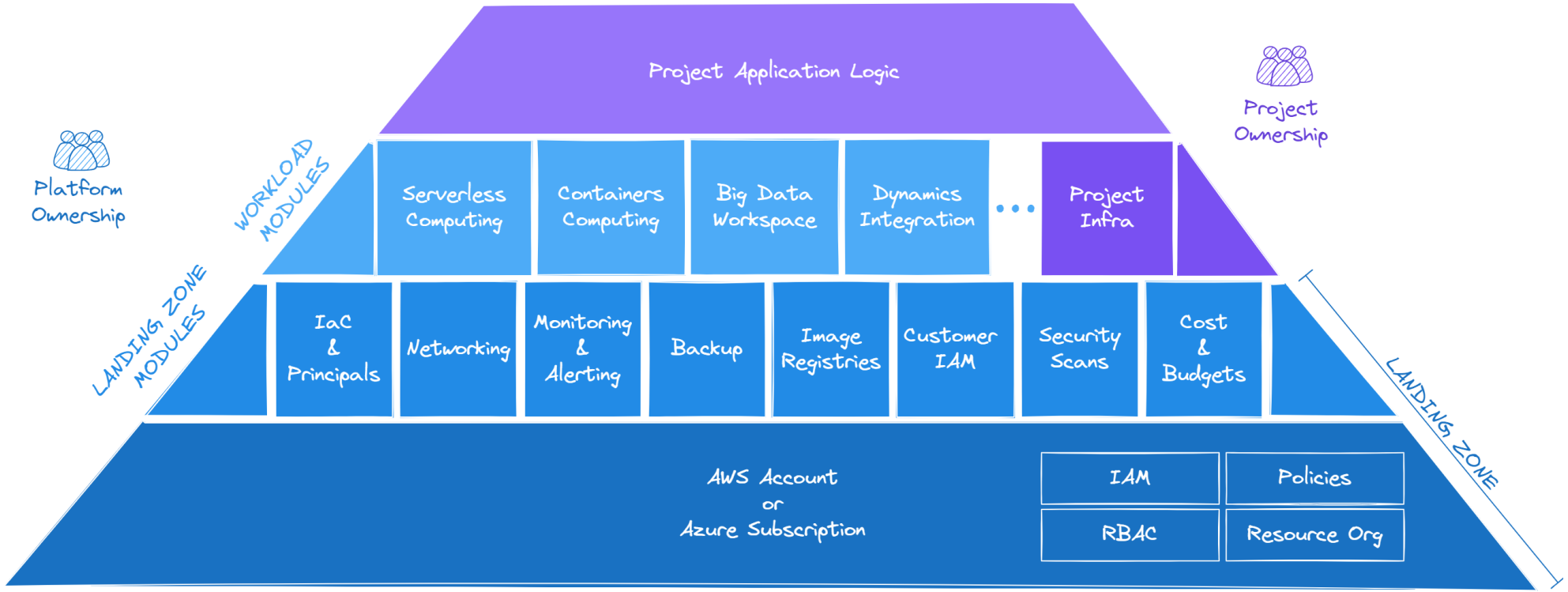
Project lands on Platform

Application

The "Goal" of our Platform!!!

This is what "lands" on



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Golden Sample – A demonstrator for usage

Golden Samples

FILTERS CI CD Angular Java NodeJS Helm Flux .NET Jenkins

Angular CI

The Angular front-end of our sample Todo Application. It demonstrates a Continuous Integration pipeline using…

Jenkins CI Angular NodeJS

CODE READ DOCS EXPLORE

Custom Jenkins Library

Custom Jenkins Library is an example of jenkins library. It demonstrates how to implements your own a jenki…

CI Jenkins

CODE READ DOCS EXPLORE

JS Vite CI

The JS Vite version of our sample Todo Application front-end. It demonstrates a Continuous Integration pipeli…

Jenkins CI Jsvite NodeJS

CODE READ DOCS EXPLORE

Monorepo Maven/NodeJS CI

Monorepo with 2 modules (backend and frontend) of a Spring Boot app of our sample Todo Application and a…

Jenkins CI Java Springboot Typescript Angular Maven

CODE READ DOCS EXPLORE

.NET Core CI

Demonstrates a Continuous Integration pipeline using ELCA Jenkins for building and testing a .NET Core application

Jenkins CI .NET

Spring Boot CI

Spring Boot back-end of our sample Todo Application. It demonstrates a Continuous Integration pipeline using…

Jenkins CI Java Springboot

GitOps CD with Flux

Let's put it all together and use Flux CD to deploy different variations of our TODO App with GitOps style declarative…

CD Gitops Flux

Helm Chart CD

See how you can use Helm and Jenkins to package a 3-Tier App as a Helm Chart and upload it to artifactory. You c…

CD Jenkins Helm

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Golden Sample – A real running app

COMPONENT — SERVICE

elca-it-sol-ecs-devops/java-spring-boot ☆

Owner DevOps team

Lifecycle production

OVERVIEW | CI/CD | API | DEPENDENCIES | DOCS | KUBERNETES

Code Quality

Not computed

0 A	2 D	4 A	0.0% E
Bugs	Vulnerabilities	Code Smells	Hotspots Reviewed

— Coverage

0.0% Duplications

Last analyzed on Aug 15, 2024, 10:57 AM

View more →

elca-it-sol-ecs-devops/java-spring-boot

Filter

ID	MESSAGE	SOURCE	WORKFLOW	STATUS	ACTIONS
1165825591 5	add prefix to attempt support of sonarqube AND sonarcloud	main 84e8fe4499602b655b1f06f9c9db02541cc38816	maven in /. - Update #911014646	Completed	
1164691762 4	add prefix to attempt support of sonarqube AND sonarcloud	main 84e8fe4499602b655b1f06f9c9db02541cc38816	Clean up container images	Error	

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Landing Zone Modules as Software Templates

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Self Service

Management

Documentation

Golden Samples

Catalog

SW Templates

Announcements

Search

Administration

Settings

Create a new component

Create new software components using standard templates in your organization

REGISTER EXISTING COMPONENT

SUPPORT

Search

PERSONAL

Starred 0

ELCA

All 2

CATEGORIES

TAGS

OWNER

Templates

service

AWS software template - container

Basic application using the AWS container blueprint. See AWS documentation for more details. v20241028.

software-template

DevOps team

CHOOSE

service

AWS software template - VPC only

IaC and GitHub workflows to deploy only a VPC. See the AWS documentations for more details. v20241028.

software-template

DevOps team

CHOOSE

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It works !

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Services
admin / Services

Self Service
Management
Documentation
Golden Sampl...
Catalog
SW Templates
Announceme...
Search
Administration
Settings

artifactory-parent
artifactory-repository
aws-account
aws-ou
azure
bitbucket
docker-hub
docker-hub-repo
flux-tenant
github
jenkins
okd-namespace
okd-project

Filters (0)
Filter

NAME	TYPE	PROJECT TEAM	CREATED AT	STATUS	ACTIONS
	bitbucket		1/1/2020	Active	
	bitbucket		7/17/2024	Archived	
	bitbucket			ve	
	bitbucket			ve	
	bitbucket			ve	
	bitbucket		7/17/2024	Archived	
	bitbucket		2/7/2023	Active	
	bitbucket		7/17/2024	Archived	
	bitbucket		1/1/2020	Archived	
	bitbucket		1/1/2020	Active	

4059

10 rows
1-10 of 4059

Key Takeaways

- Status
  - 95% transition from VM to container for on-prem's
  - 90% of project CI/CD adoption
  - Project SonarQube adoption with KPI reporting up to the management

- Challenges
  - Impossible to cover all needs -> 80% / 20% rule -> product management
  - Need skills, Platform Eng. team do everything (Build + Run + Support)
  - Highly transversal , need alignment on goals, terminology, communication
  - Testing is hard, nothing is similar to prod (WireMock is a friend)

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Thank you

Q&A

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